

# ADVANCED LINEAR ALGEBRA HOMEWORK 6

ROB TOMPKINS

**P.4.9.** If  $A \in \mathbf{M}_n$  is to be represented as  $A = XY^\top$  for some  $X, Y \in \mathbf{M}_{n \times r}$ , explain why  $r$  cannot be smaller than  $\text{rank}(A)$ .

*Solution.* We claim that the columns of  $A$  must lie in the column space of  $X$ . To see this we write

$$Y^\top = \begin{bmatrix} y_1^\top \\ y_2^\top \\ \vdots \\ y_r^\top \end{bmatrix},$$

where  $y_i \in \mathbb{F}^n$  are columns of  $Y$ . Then we have that

$$XY^\top = \sum_{k=1}^r x_k y_k^\top.$$

Now choose an arbitrary vector  $v \in \mathbb{F}^n$ ,

$$Av = XY^\top v = X(Y^\top v)$$

with  $Y^\top v \in \mathbb{F}^r$ . So  $Av$  is a linear combination of the  $r$  columns of  $X$ , which gives us  $\text{col}(A) \subseteq \text{col}(X)$ . So

$$\text{rank}(A) = \dim(\text{col}(A)) \leq \dim(\text{col}(X)) = \text{rank}(X) \leq r$$

So  $\text{rank}(A) \leq r$ , and we are done. □

**P.4.10.** For  $X \in \mathbf{M}_{m \times n}$ ,  $\nu(X)$  denote the nullity of  $X$ .

- (a) Show that  $\nu(X^\top) = \nu(X) + m - n$ .
- (b) Let  $A \in \mathbf{M}_{m \times k}$  and  $B \in \mathbf{M}_{k \times n}$ . Show that the rank inequality given by

$$\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$$

is equivalent to

$$(1) \quad \nu(AB) \leq \nu(A) + \nu(B).$$

(c) If  $A, B \in \mathbf{M}_n$ , show that the rank inequality given by

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

is equivalent to

$$(2) \quad \max\{\nu(A), \nu(B)\} \leq \nu(AB)$$

The inequalities (??) and (??) are known as *Sylvester's law of nullity*.

(d) Consider  $A = \begin{bmatrix} 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$ . Show that the inequality (??) need not be valid if the matrices are not square.

*Proof.* (a) By the rank-nullity theorem

$$\begin{aligned} \text{rank}(X) + \nu(X) &= n \\ \text{rank}(X) &= n - \nu(X) \\ \text{rank}(X^\top) + \nu(X^\top) &= m \\ \text{rank}(X^\top) &= m - \nu(X^\top) \end{aligned}$$

Yet we know that  $\text{rank}(X) = \text{rank}(X^\top)$ . So we have:

$$\nu(X^\top) = \nu(X) + m - n.$$

(b) Similar to (a) we write out the rank equations for the matrices

$$\begin{aligned} \text{rank}(A) &= k - \nu(A) \\ (3) \quad \text{rank}(B) &= n - \nu(B) \\ \text{rank}(AB) &= n - \nu(AB). \end{aligned}$$

Substituting into  $\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$  we get

$$\begin{aligned} \text{rank}(A) + \text{rank}(B) - k &\leq \text{rank}(AB) \\ (k - \nu(A)) + (n - \nu(B)) - k &\leq n - \nu(AB) \\ n - \nu(A) - \nu(B) &\leq n - \nu(AB) \\ \nu(A) + \nu(B) &\geq \nu(AB), \end{aligned}$$

where in that last step we both subtracted both sides by  $n$  and multiplied both sides of the inequality by  $-1$  switching its direction. And we are done.

(c) Similarly to (b) let's consider equation (??). If we consider what  $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$  is actually saying we have that we have both  $\text{rank}(AB) \leq \text{rank}(A)$  and  $\text{rank}(AB) \leq \text{rank}(B)$ .

We see that

$$\begin{aligned} n - \nu(AB) \leq n - \nu(A) &\iff \nu(A) \leq \nu(AB), \\ n - \nu(AB) \leq n - \nu(B) &\iff \nu(B) \leq \nu(AB). \end{aligned}$$

So we have

$$\max\{\nu(A), \nu(B)\} \leq \nu(AB)$$

(d) We compute  $AB$  and  $BA$  for the sake of discussion

$$AB = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \quad \text{and} \quad BA = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

We also look at what the null spaces of the respective matrices are. So we describe our linear transformations as functions

$$A = \begin{bmatrix} 0 & 1 \end{bmatrix} = (x, y) \mapsto y \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \hat{x} \mapsto \begin{bmatrix} \hat{x} & \hat{0} \end{bmatrix},$$

where  $\hat{x} = [x_1 \ \cdots \ x_n]$  and  $\hat{0}$  is the  $n$ -ary zero vector. Given that  $\ker(A) = \{(x, 0) : x \in \mathbb{F}\}$  we have that  $\nu(A) = 1$ . Also, notice  $\ker(B) = \{\hat{0}\}$ , so  $\nu(B) = 0$ . Lastly notice that both  $\nu(AB) = 1$  by  $AB$  being the zero map, and  $\nu(BA) = 1$  by single element linear transformation. But then we have  $\max\{0, 1\} \leq 1$  in either case which does not help us. Hm.

So we consider the following counterexample

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

So,

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The nullity of  $A$  is 1 because our rank is 2. The nullity of  $B$  is 0, and finally the nullity of  $AB$  is 0 because it's invertible. So we truly are stuck with  $\max\{0, 1\} \leq 0$  which doesn't hold.  $\square$

**P.4.13.** Let  $\mathcal{V}$  be a nonzero  $n$ -dimensional vector space over the field  $\mathbb{F}$ , and let  $T \in \mathfrak{L}(\mathcal{V}, \mathbb{F})$  be a nonzero linear transformation. Why is  $\dim(\ker(T)) = n - 1$ ?

*Proof.* Let  $T : \rightarrow \mathbb{F}$  be a linear transformation. Now the  $\text{rank}(T) \leq 1$ , and because  $T$  is nonzero, we must have  $\text{rank}(T) = 1$ . By the rank nullity theorem we see that

$$\begin{aligned} 1 + \nu(T) &= n \\ \nu(T) &= n - 1. \end{aligned}$$

Done. □

**P.4.35.** For each of the following matrices, compute its reduced row echelon form, its pivot column decomposition, and bases for its column space and null space.

$$(a) \begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}.$$

*Solution.* (a) We begin by performing Gauss-Jordan elimination on the matrix without augmentation.

$$\begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix}$$

$$r_1 : -r_1$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix}$$

$$r_2 : r_2 - 3r_1, \text{ and } r_3 : r_3 - 2r_1$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 7 & 14 & 8 \\ 0 & 0 & 3 & 6 & 8 \end{bmatrix}$$

$$r_2 : r_2/7$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 3 & 6 & 8 \end{bmatrix}$$

$$r_3 : r_3 - 3r_2$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 0 & 0 & \frac{32}{7} \end{bmatrix}$$

$$r_3 : 7/32r_3$$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & -3 \\ 0 & 0 & 1 & 2 & \frac{8}{7} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$r_1 : r_1 + 3r_3$ , and  $r_2 : r_2 - (8/7)r_3$

$$\begin{bmatrix} 1 & 2 & -2 & -3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Finish with  $r_1 : r_1 + 2r_2$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns indices are 1, 3, 5. We can write  $2a_1 = a_2$  and  $a_4 = a_1 + 2a_3$ . We take our basis for  $\text{col}(A)$  as the original columns in our pivot column indices as

$$\left\{ \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

For the null space, we need to solve  $Ax = 0$ . Letting  $x = [x_i]$  with  $i = 1, \dots, 5$ , we expand to get

$$x_1 + 2x_2 + x_4 = 0$$

$$x_3 + 2x_4 = 0$$

$$x_5 = 0$$

For simplicity, we remove subscripts as  $x_2 = y$  and  $x_4 = z$  which gives us

$$x_1 = -2y - z, \quad x_3 = -2z, \quad x_5 = 0$$

This gives us a basis for our null space as:

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

(b) We begin by again using Gauss-Jordan elimination for our row-reduction algorithm. The computation follows:

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}$$

$r_2 : r_2 - r_1$  and  $r_3 : r_3 - 3r_1$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 & -2 \\ 0 & 5 & -4 & 0 & -7 \end{bmatrix}$$

$r_3 : r_3 - 5r_2$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

$r_2 : r_2 + r_3$

$$\begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}$$

Finally, we perform  $r_1 : r_1 - 2r_3$ , and  $r_1 : r_1 + r_2$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \end{bmatrix}.$$

The pivot columns here are 1, 2, 3. So our basis from our original matrix follows as:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

To get a basis for the nullity of  $B$  we expand  $Bx = 0$ , giving us the following equations

$$x_1 + x_4 - 2x_5 = 0$$

$$x_2 + x_5 = 0x_3 + 3x_5 = 0$$

So  $x_1 = -x_4 + 2x_5$ ,  $x_2 = x_5$ , and  $x_3 = -3x_5$ . Then we let  $x_4$  and  $x_5$  range freely to see that the basis for our null space is

$$\left\{ \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

□

**P.4.38.** Suppose that  $A \in \mathbf{M}_{6 \times 4}$  is row equivalent to

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find a basis for  $\text{null}(A)$ .
- (b) What is  $\dim(\text{null}(A^\top))$ ?

*Solution.* (a) We first find the  $\text{rank}(B)$ . We do this by doing row-reduction on  $B^\top$ , which follows as

$$B^\top = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 0 & 0 & 0 \end{bmatrix}$$

$$r_4 : r_4 - r_1, \quad r_2 : 1/2r_2, \quad r_3 : r_3 - 3/2r_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \end{bmatrix}$$

$$r_3 \leftrightarrow r_4$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r_3 : r_3 - 2r_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r_3 : 1/4r_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

So we have  $\text{rank}(B^\top) = 3 = \text{rank}(B)$ , and  $\text{rank}(B) + \nu(B) = 4$ . Thus, we have that  $\nu(B) = 1$ . A basis for  $\nu(A)$  would be  $e_4$  as our row-reduction above contains  $e_1, e_2, e_3$ .

(b) Similarly to in (a)  $\nu(A^\top) = 1$  which is the dimension of the nullity.

□